# 1.0 Executive Summary

This proposal outlines the initial research, design, fabrication, and testing for the University of Pennsylvania DBF Team in pursuit of the 2026 AIAA Design/Build/Fly Competition in Wichita, KS. The 30th annual competition centers around the Banner Towing Bush Plane, including one ground mission and three flight missions: Test Flight, Charter Flight with passengers and cargo, and a Banner Flight with in-air deployment. Scoring is determined by lap time, number of passengers/cargo flown, banner length, and successful landing of the plane.

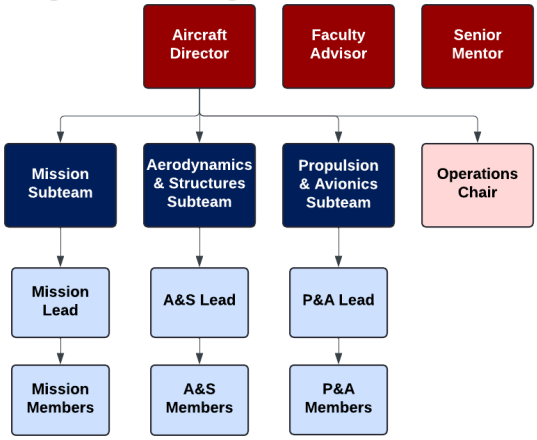
This year's aircraft is a high-wing, single-propeller, taildragger. By utilizing Computational Fluid Dynamics, sensitivity analysis, and flight simulations conducted using a flight computer (testing purposes only) and QGround Control, the aircraft configuration aims to combine simple takeoff and landing, stable and efficient flight, and a high maximum speed to achieve and exceed the required mission goals. The team has developed a comprehensive manufacturing plan, outlining the next steps in design, CAD modeling, simulation, and physical testing, as detailed in the following sections.

# 2.0 Management Summary

## *2.1 Organization Summary, Team Roles, & Subteam Skills*

Penn DBF is composed of 3 subteams: Aerodynamics & Structures (A&S), Propulsion & Avionics (P&A), and Mission. A&S designs and tests the plane's mechanical structure, P&A integrates the aircraft's control and propulsion systems, and the Mission team optimizes the plane for the specific challenge of the year. In addition to subteams, Penn DBF leadership includes an Aircraft Director, an Operations Chair, a Faculty Advisor, and senior mentors who assist with major milestone reports and design reviews (Figure 2.1).

**Figure 2.1 – Organizational Chart**




The DBF Director coordinates among the subteams, ensuring the overall timeline and milestones are followed. Each week, general meetings are held to facilitate cross-subteam communication and interdisciplinary work, with subteams holding additional meetings throughout the week. Lastly, Penn DBF is split between returning members and new "Cadets." Returning members assist with onboarding and educating new members through workshops and mentorship, enabling our Cadets to quickly and effectively contribute to our team.

**Figure 2.2 – Subteam Skills**

| Subteam | Skills Required |
|---|---|
| Aerodynamics & Structures | - CAD (SolidWorks) proficiency<br>- Knowledge in FEA and CFD processes<br>- Rapid Prototyping (laser cutting, 3D printing)<br>-Design for Manufacturing |
| Propulsion & Avionics | - Knowledge of singals, electronics, wing, and propulsion systems<br>- CAD proficiency<br>- C/Arduino proficiency |
| Mission | - Knowledge of materials and material selection<br>- Multi-objective optimization<br>- Constraint analysis and management<br>- Fabrication techniques & iterative testing |

## *2.2 Budget & Major Milestone Chart*

The proposed schedule for the design and manufacture of the project is outlined in Fig. 4. The DBF Director holds the subteam leads and their teams accountable for completing the necessary milestones throughout the year. Team progress is updated regularly to ensure key deadlines are met and the team stays on track. An estimate of Penn DBF's expenses for the 2025-2026 Competition year can be found in Figure 2.3. The team estimates the total budget at approximately $5,400. A majority of this budget (approx.

$3,900) is projected to be travel expenses, such as lodging, airfare, and rental vehicles. Penn DBF's yearly budget will be provided by the Penn Engineering Student Activities Council, through the larger Penn Aerospace Club.

**Figure 2.3 - Milestone Schedule**

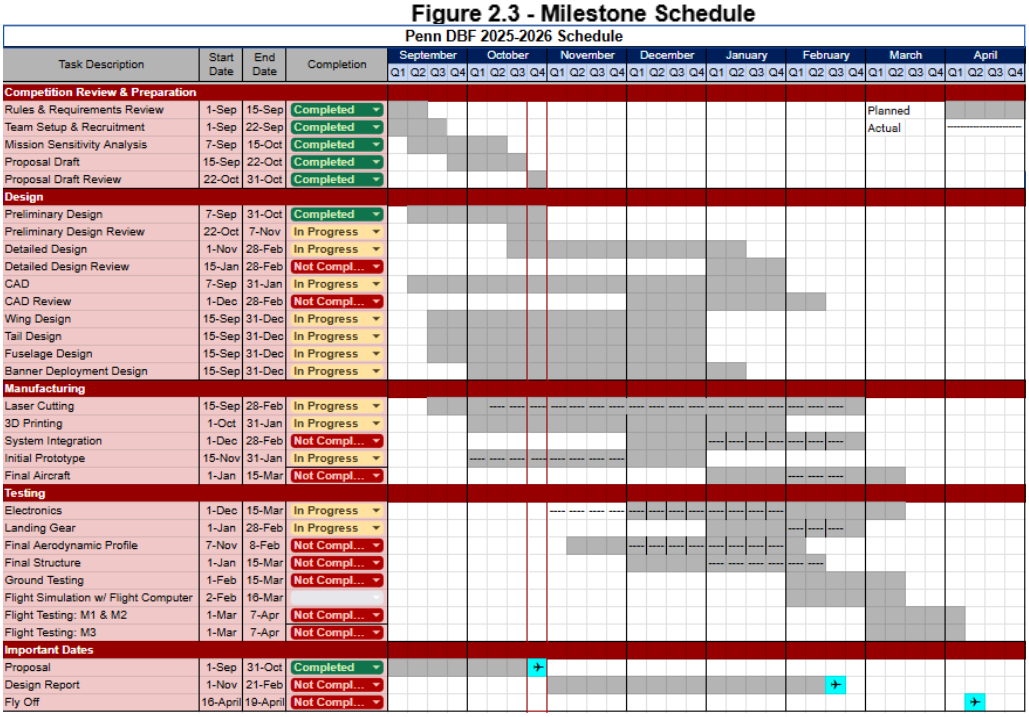



**Figure 2.4 - Budget**

| | # | Cost (USD) | Description |
|---|---|---|---|
| **Components** | | | |
| Avian 5065-450Kv Brushless Motor | 1 | $140.00 | 450Kv, 1200W, 100A |
| 25kg Servo | 2 | $30.60 | 25 kg/cm |
| 20kg Servo | 2 | $30.00 | 20 kg/cm |
| 18x12 Propeller | 1 | $25.00 | ------ |
| 22.2V 6S Lipo Battery | 1 | $60.00 | 22.2V, 4000mAh |
| 6V 2000mAh NiMH Battery | 1 | $12.00 | 6V, 2000mAh |
| Flyski Transmitter + Reciever | 1 | $64.00 | ------ |
| 100 Amp Brushless Smart ESC | 1 | $100.00 | 100A rating |
| 3" Treaded Wheels | 3 | $13.00 | ------ |
| Pixhawk 6C Flight Contoller | 1 | $212.00 | ------ |
| Miscellaneous | ------ | $50.00 | Fuses, switches, etc. |
| **Materials** | | | |
| MDF Board (0.25" Thickness) | 30 | $200.00 | 18" x 24" sheets |
| Bass wood | 10 | $80.00 | 1/8" Sheets |
| Carbon Fiber Rods, 1" OD | 3 | $120.00 | 4' rods |
| Foam Sheets | ------ | $50.00 | Rolled sheets, 1/8" thick |
| Shrink Wrap | 2 | $40.00 | Two Rolls |
| Miscellaneous | ------ | $300.00 | Fasteners, Tools, Supplies |
| **Operations** | | | |
| AIAA Memberships | 21 | $777.00 | $37/membership, 21 students |
| Airfare, Round-trip | 6 | $2,400.00 | $200/flight/person, 6 students |
| Lodging | ------ | $500.00 | 4 nights, 6 students |
| Rental Car | 1 | $250.00 | 1 rental car, 4 days |
| **Estimated total cost** | | $5,453.60 | |

# 3.0 Conceptual Design Approach

## *3.1 Decomposition of Mission Requirements*

**Figure 3.1: Mission Requirements**

| Mission | Scoring | Description | Subteam Requirements |
|---|---|---|---|
| M1: Delivery Flight | $M1 = 1.0$ for successful mission | - Baseline flight (no payload)<br>- Complete 3 laps within 5 minutes w/ successful landing.<br>- Validates flight stability, control authority, and propulsion reliability | - Ensure stable and controlled flight.<br>- Responsive control surfaces and throttle.<br>- Verify aerodynamic and propulsion performance in clean configuration. |
| M2: Charter Flight | $M2 = 1 + \left(\frac{N_{net\ income}}{Max_{net\ income}}\right)$ | - Carry passengers (2.3" rubber ducks) and cargo (3" hockey pucks) to maximize net income.<br>- Must carry ≥ 3 passengers per cargo.<br>- Maximize laps in a 5-minute flight window.<br>- Net income = Income − Cost, based on payload, laps, and battery capacity. | - Optimize number of passengers/cargo for best net income.<br>- Secure all payloads for flight.<br>- Maximize completed laps for revenue.<br>- Minimize cost via efficient battery selection. |
| M3: Banner Flight | $M3 = 2 + \left(\frac{N_{\frac{\#laps \times banner\ length}{RAC}}}{Max_{\frac{\#laps \times banner\ length}{RAC}}}\right)$ | - Tow and release banner ≥10 ft long, AR ≤ 5:1.<br>- Deploy banner after upwind turn & release after finish line.<br>- 5-minute flight window; must successfully land to receive a score. | - Reliable deploy/release system for banner.<br>- Maintain stability and control with added drag.<br>- Design banner for lightweight, low-drag towing.<br>- Ensure quick release to minimize retrieval time. |
| Ground Mission | $GM = \frac{\text{Min}_{\text{mission time}}}{N_{\text{mission time}}}$ | - Timed ground assembly<br>- Demonstrate flight controls, unload, install banner, and perform a banner deploy/release demonstration. | - Simple assembly procedure<br>- Accessible hatches and mounts for efficient install/removal.<br>- Smooth crew coordination - Verify flight control checks and banner deployment reliability. |
| Total Mission Score = M1 + M2 + M3 + GM | | | |

This year's mission sequence tests the aircraft's stability, efficiency, and operational readiness across flight and ground conditions. Mission 1 establishes baseline performance by completing three laps in five minutes under no-payload conditions, verifying stability, control, and propulsion reliability. Mission 2 introduces passengers (2.3 in ducks) and cargo (3 in pucks) to maximize net income within the 5-minute window, requiring at least three passengers per cargo and balancing payload layout, drag, and energy efficiency under the 100 W-hr power limit. Mission 3 evaluates banner towing and release, scored by (#laps x bannerlength)/RAC, with RAC = 0.05 × wingspan + 0.75; the banner (≥ 10 in, AR ≤ 5:1) must stow, deploy, and release reliably without compromising control, introducing a trade-off between wingspan and drag. The Ground Mission measures loading and deployment speed in a 10 ft × 10 ft area, emphasizing quick assembly, accessibility, and secure payload handling. Together, these missions guide subsystem requirements across aerodynamics, propulsion, structure, and payload integration to achieve an efficient, reliable aircraft.

## *3.2 Sensitivity Analysis*

Based on the scoring formulas for this year's M2 and M3, we conducted a sensitivity analysis (Figures 3.2 and 3.3) on various aircraft parameters and their corresponding impacts on M2 and M3 scores. As a result, we have identified net income and banner length-to-weight ratio as the most influential factors.

- **Net Income:** Maximize revenue per passenger and unit of cargo while maximizing airspeed and minimizing drag
- **Banner Length to Weight**: Maximize banner length while accounting for drag

**Figure 3.2 – M2 Score Sensitivity Analysis**

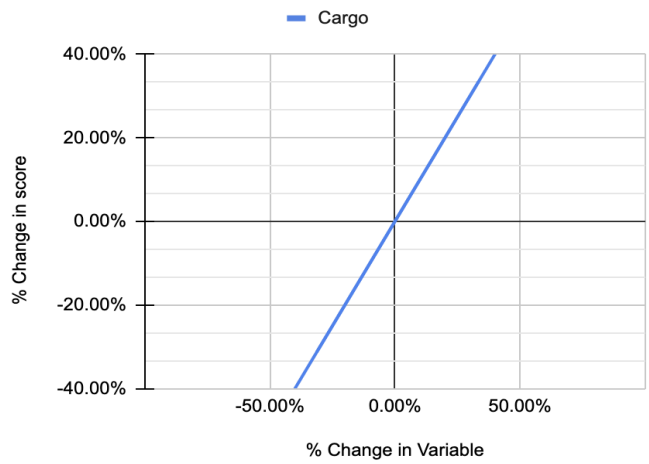
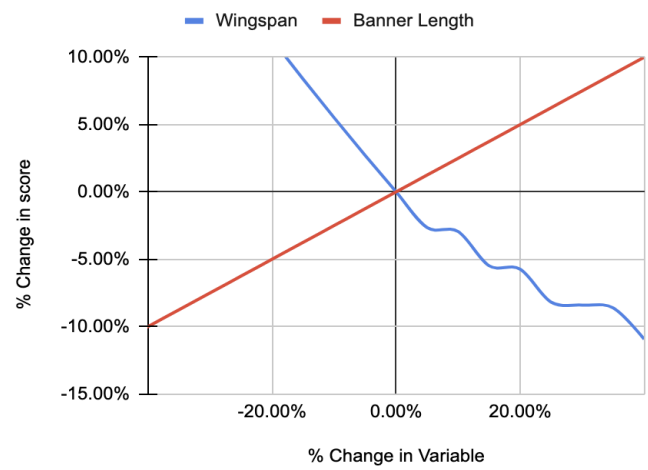


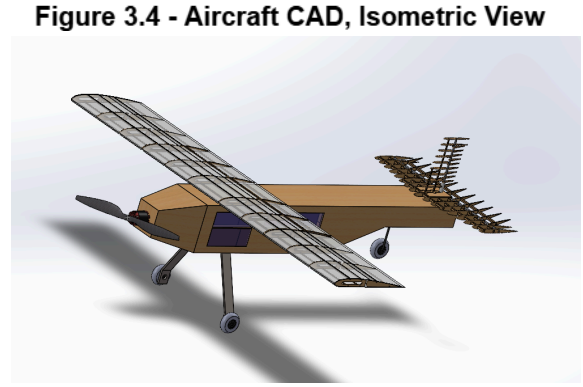

**Figure 3.3 – M3 Score Sensitivity Analysis**




## *3.3 Preliminary Design & Concept Sketches*

The plane's design was done with a focus in two categories, the fuselage for mission optimization, and the wing and tail for flight control, stability, and speed. In order to maximize cargo space and ensure stability during banner towing, a conventional fuselage design with a main square cross section area of 25 in$^2$ was selected. This sizing allows the fuselage to contain two vertically aligned compartments with one row ducks and stacks of two pucks respectively, each with a door for easy access during the ground mission. The walls of our fuselage will be a truss design

**Figure 3.4 - Aircraft CAD, Isometric View**



wrapped in heat-shrink covering to maintain aerodynamic integrity. Our team will conduct research on truss designs and run FEA on structural members in order to minimize weight while maintaining stability.

For our wing, a trade study was conducted for three wing placements and three tail designs as seen in Figures 3.5 and 3.6. The team chose a high-wing design, based on its superior stability and lift-to-drag ratio, with a conventional tail design based on its high ease of build and strong stability. Our wing will use the E423 airfoil which provides high lift at low Reynolds numbers while being easy to manufacture, with a carbon fiber rod run through the wing to ensure structural integrity and flexibility. This design ensures a high strength to weight ratio. The wing has an untapered span of 60” and a chord length of 12” (AR = 5). Both the horizontal and vertical stabilizers will use the NACA 0010 symmetric airfoil. The horizontal stabilizer will have a span of 12” and a chord length of 8” while the vertical stabilizer will have a chord length of 8” and a height of 8”. The horizontal stabilizer will be connected via the vertical stabilizer, which runs through the back of the fuselage for perpendicularity and security during flight. Figure 3.5 shows an XFLR5 simulation of the plane accounting for the masses of main electronic components, cargo, and passengers. Figure 3.6 shows that the Cm vs. Alpha is downward sloping with a slope of around -0.57 $rad^{-1}$, indicating that the plane is stable in the direction of its length.

**Figure 3.5 - XFLR5 Wing & Tail Simulation**

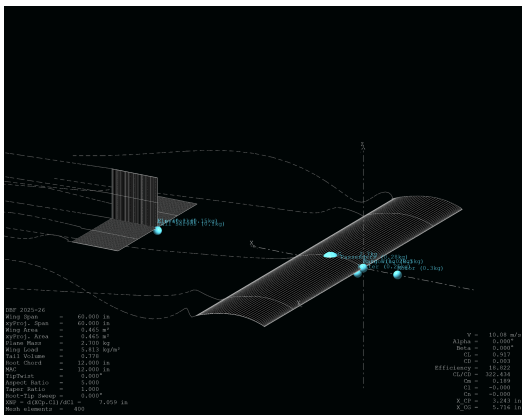
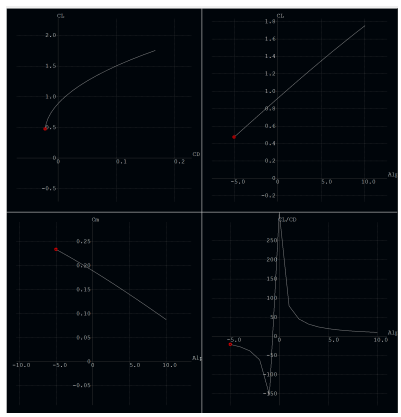


**Figure 3.6 - XFLR5 Simulation Data**



Our tractor, single-prop configuration, will put less stress on our propeller while keeping efficient use of weight and space. We adopted a taildragger configuration for the landing gear due to its capability to provide a greater angle of attack during takeoff. To power our motor, a 6S LiPo battery with 4000mAh capacity that stays below 100 Wh limit will be used, leaving us with a flight time of approximately 4.4 minutes. Our propulsion system will be powered by the Avian 5065-450Kv Outrunner Brushless Motor with a 17” x 10” propeller, giving the plane a 1.52 thrust-to-weight ratio, and an estimated flight speed of 10 m/s. Aircraft specifications were found using eCalc. Detailed avionics specs can be found in Figure 3.7.

**Figure 3.7 - Electronics Selection**

| Part | Description |
|---|---|
| BETU 20 KG and 25 KG Servos | Two 25 kg servos for ailerons, a 20 kg servo elevator, and a 20 kg servo for rudder. |
| Tenergy NiMH 2000mAh 6V Servo Battery | We chose a 6 V, 2000 mAh battery to power all four servos. Its compact size and capacity are well-suited to our needs. |
| Avian 5065-450Kv Outrunner Brushless Motor | Paired with our 6s 22.2 V under ideal conditions should achieve 9094 RPM [eCalc]. |
| YOWOO 22.2V 6S Lipo Battery 4000mAh 100C | To comply with the 100 Wh limit, we chose a 6S LiPo battery with a 4000 mAh capacity. |
| Flysky FS-i6X 6-10 RC Transmitter, FS-iA6B Receiver | Reliable 2.4GHz AFHDS technology minimizes interference and ensures stable communication. |
| Avian 100 Amp Brushless Smart ESC | 100 A ESC for its smart all-in-one telemetry and automatic LiPo low-voltage cutoff. |

**Figure 3.8 – Tail Design Decision Matrix**

| Figure of Merit | Conventional | T-Tail | H-Tail |
|---|---|---|---|
| Drag | 3 | 2 | 1 |
| Ease of Build | 5 | 3 | 2 |
| Maneuverability | 3 | 4 | 5 |
| Stability | 4 | 4 | 5 |
| Weight | 4 | 4 | 3 |
| **Total** | 19 | 17 | 16 |

**Figure 3.9 - Wing Position Decision Matrix**

| Figures of Merit | Low Wing | Mid Wing | High Wing |
|---|---|---|---|
| Takeoff | Confirmed | Confirmed | Confirmed |
| Aerodynamics | 8 | 5 | 6 |
| Stability | 7 | 8 | 8 |
| Lift to Drag Ratio | 8 | 9 | 10 |
| Total | 21 | 22 | 24 |

The fuselage interior was designed for optimal space efficiency for both passenger (Fig 3.10) and cargo (Fig 3.11) layouts. To minimize ground handling time and ensure in-flight security, polyethylene foam block inserts with press-fit cutouts were designed. Three materials for the blocks were considered: polystyrene, polyurethane, and polyethylene. Polystyrene, while cheap, is fragile and difficult to shape effectively. We found Polyurethane to be excessively compressable, at a coefficient of about 40%, making inserts liable to slip during loading and flight. Polyethylene, though higher in cost, is lightweight, easy to shape, and sufficiently stable. Therefore, both passenger and cargo configurations utilize polyethylene foam blocks and press-fit interfaces for the restraint system. After testing, expected time for the ground mission is around 14 seconds.

**Figure 3.10 - Passenger Block**

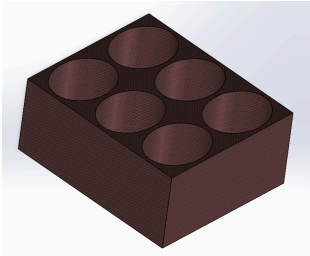
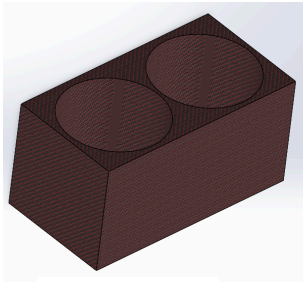


**Figure 3.11 - Cargo Block**

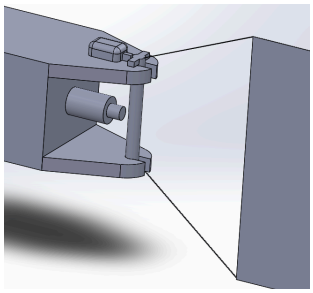


The passenger compartment includes a 6.9 x 6.5 x 3in polyethylene block accommodating 18 passengers. Each block features 2.1 x 2.7in oval press-fit cutouts that restrain the passengers. The block itself is press-fit into the fuselage and can be removed through two side access doors for simple loading and unloading. The cargo compartment, accessible via a separate side door, houses two 3.5 × 6.5 x 3in polyethylene blocks with four 3.1in diameter cutouts, providing a secure fit for the cargo.

### *3.4 Banner Deployment and Release Mechanism*

Penn DBF's banner will be stored directly behind the tail of the plane on a horizontal roller connected to a clamping mechanism. The vertical roller will use a stopper servo to allow the roller to unwind and banner to fly freely with two attachment points located at the bearing joints of the roller to prevent tangling upon release. There is to be a hole in the mount with a horizontal split keeping the roller bearings in place which, with the horizontal force from a solenoid, can be pushed open and the banner released from the plane, resulting in a smooth drop and retrieval process.

**Figure 3.12 - Banner Release Mechanism**

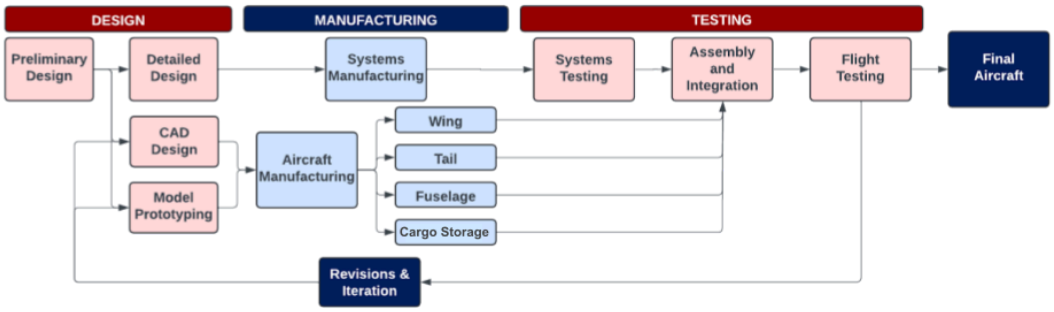


## 4.0 Manufacturing Plan

### *4.1 Preliminary Manufacturing Flow*

The aircraft design iteration process shown in Figure 4.1 illustrates the stages of design, manufacturing, testing, and revisions. Initial proof of concepts are made in SolidWorks, and then prototypes will be constructed. Following ground function and flight testing, revisions will be made to revise designs until function. The design process will be repeated as needed. Aircraft manufacturing will involve several fabrication methods and materials depending on part requirements. Primary fabrication methods include 3-D printing and laser-cutting.

**Figure 4.1 – Preliminary Manufacturing Flow**



### *4.2 Critical Manufacturing Process & Technologies*

While selecting materials for the aircraft, we considered material strength and weight for balsa wood, bass wood, and Medium-Density Fiberboard (MDF). We decided on basswood which has a higher strength to weight ratio compared to MDF and, although it has a less optimal strength to weight ratio compared to balsa, the strength of our motor warrants a more durable structure which basswood provides while still allowing for flight capable speeds. We will use carbon fiber rods to support critical areas of the aircraft, like the wings and tail as they are lightweight and can sustain great forces in tension. The wings and tail will be made using the rib-and-spar method. After the main structure is completed, propulsion and avionics systems (Figure 3.6) will be installed in the aircraft. The main aircraft structure will be a truss design and wrapped in a heat-shrink film to increase structural support.

## 5.0 Test Planning

Figure 5.1 outlines Penn DBF's testing plan leading up to the competition Fly-Off. Each subteam is assigned specific tests for which they are responsible. If a test cannot be completed by its deadline, this must be communicated to the Team Director and Subteam Leads to maintain alignment with the overall testing plan and project timeline outlined in Section 2. All team members are required to attend flight tests. This year, DBF is expanding its testing plan to include 'Dummy Flights' (or simulated flights) using a Pixhawk flight computer and QGroundControl ground station. These simulations will collect data on control surface response, throttle behavior, and overall system performance prior to live flight testing. Using the data, we can create graphs and analyze the flight to help decision-making and design choices in the future. The flight computer integration into the plane's avionics is for test purposes only, as it is not allowed to have a flight computer during competition.

**Figure 5.1 – Preliminary Test Plan**

| Category | Test | Team In Charge | Equipment Needed | Objective | Deadline |
|---|---|---|---|---|---|
| Ground | Control Surfaces Test | Propulsion & Avionics | All Control Surface assemblies (including servo motors, linkages, and mounts) | Validate that control surfaces have proper range-of-motion in both directions | Q2 Dec |
| Ground | Propulsion System Test | Propulsion & Avionics | Motor, propeller, and motor wiring/connections; Motor mounting (i.e. testing stand) | Validate that propulsion system can run for all possible operating speeds without error | Q2 Dec |
| Ground | Landing Gear Test | Aerodynamics & Structures | Landing gear mounts and wheels; weights | Validate that landing gear is able to withstand maximum possible impact during takeoff and landing; derive aircraft weight from CAD model | Q2 Dec |
| Ground | Wing Structure/Strength Test | Aerodynamics & Structures | Physical wing assembly derived from CAD and manufacturing plan; hanging weights | Validate that wing assembly has sufficient strength during worst conditions of flight (i.e. maximum wing stress from aerodynamic | Q4 Jan |
| Ground | Wing Aerodynamics Test | Aerodynamics & Structures | Phyiscal wing assembly derived from CAD and manufacturing plan; SEAS wind tunnel | Validate that wing assembly has desired flight properties (drag and lift) via wind tunnel data; compare to CFD calculations | Q4 Jan |
| Ground | Fuselage Structure/Strength Test | Aerodynamics & Structures | Physical fuselage assembly derived from CAD and manufacturing plan; hanging weights; mission components | Validate that wing assembly has sufficient strength without failure during maximum fuselage load, including mission components (i.e. banner assembly, duck storage) | Q2 Feb |
| Ground | Tail Structure/Strength Test | Aerodynamics & Structures | Physical tail assembly derived from CAD and manufacturing plan; hanging weights | Validate that tail assembly has sufficient strength during worst conditions of flight (i.e. maximum tail stress from aerodynamic forces) | Q2 Feb |
| Ground | Ground Mission Test | Mission | Ground Mission Checklist; Mission components (Banner, ducks, and hockey pucks) | Ensure correct operation of Ground Mission as outlined in Competiton Rules | Q2 March |
| Flight | Simulated flight with Flight Computer | Propulsion & Avionics | Flight Computer; QGroundControl (ground station) | Simulate a flight and gather data in flight comuter to analyze . | Q2 Feb |
| Flight | Initial Flight Test | **All Teams** | Complete Aircraft Prototype | Determine flight constraints and design/manufacturing challenges for current prototype) | Q1 March |
| Flight | Mission Flight Test | **All Teams** | Complete Aircraft Prototype, Finalized Mission designs and sequences | Test aircraft with individual mission requirments and parameters | Q1 Apr |
| Flight | Simulated Competition | **All Teams** | Finalized Aircraft | Simulate full competition with ground mission, flight missions, and competition timing. Ensure full aircraft fluidity during flight | Q1 Apr |